Dear Hiring Manager,

I am writing to express my interest in the Analyst – Data Science position within the Model Risk Management Group at American Express. As a recent MSc Computer Science graduate from Sapienza University of Rome, specializing in Artificial Intelligence and Machine Learning, I am excited by the opportunity to contribute to the governance, evaluation, and responsible deployment of Generative AI and advanced machine learning models.

Throughout my master's program, I developed a strong foundation in AI research, model evaluation, and statistical analysis. My thesis, “Evaluating Pretrained Vision Transformers Under Continual Learning”, involved designing reproducible experimental pipelines, benchmarking transformer-based models, analyzing robustness under changing data distributions, and evaluating model reliability and catastrophic forgetting using Python and PyTorch. This experience strengthened my ability to critically assess model performance and communicate technical findings through structured research.

In addition to my research, I built several AI applications that strengthened my understanding of modern machine learning systems. My DocDelta project leveraged Sentence Transformers, FastAPI, and AWS to perform semantic document analysis and comparison, providing practical experience with transformer-based models, embeddings, and retrieval-oriented workflows. I also explored semantic text compression using transformer embeddings, evaluating reconstruction quality through quantitative metrics and performance analysis. These projects have given me hands-on exposure to technologies that underpin today's Generative AI systems.

I am particularly drawn to this role because of its emphasis on independent model assessment, AI governance, and responsible AI practices. I enjoy understanding not only how models are built, but also how they should be evaluated for robustness, bias, explainability, and real-world reliability. The opportunity to support model risk reviews, contribute to AI governance frameworks, and stay at the forefront of Generative AI aligns closely with both my academic background and my long-term career aspirations.

Beyond my technical skills in Python, SQL, PyTorch, TensorFlow, and machine learning, I bring a research-oriented mindset, strong analytical thinking, and the ability to communicate complex technical concepts clearly. Having completed research, published an IEEE conference paper, and collaborated on multiple AI projects, I am eager to contribute these skills while continuing to learn from the experienced team at American Express.

I would welcome the opportunity to discuss how my academic research, technical background, and enthusiasm for responsible AI can contribute to the Model Risk Management Group. Thank you for your time and consideration. I look forward to hearing from you.

Sincerely,

Swati Aggrawal